

Where & How Windows Stores & Encrypts Passwords

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1. Windows Login Passwords (Local Accounts)

Stored in: C:\Windows\System32\config\SAM What's inside: - NT password hash - (Legacy) LM hash - User SIDs - Account flags Protection: - SAM content encrypted with the SAM Key - SAM Key encrypted with the BootKey BootKey stored in: C:\Windows\System32\config\SYSTEM
Architecture: SYSTEM hive → BootKey → decrypts SAM → SAM Key → decrypts NT hash

2. Browser Passwords (Chrome / Edge)

Stored in: C:\Users\<Username>\AppData\Local\Google\Chrome\User Data\Default>Login Data
Protection: - Passwords encrypted using DPAPI - DPAPI is tied to the Windows login password Meaning: the password database is useless outside your Windows account.

3. Wi-Fi Passwords

Stored in: C:\ProgramData\Microsoft\Wlansvc\Profiles\Interfaces\ Protection: - Wi-Fi key (keyMaterial) encrypted with DPAPI.

4. DPAPI (Data Protection API)

Purpose: - Used by Windows and apps to encrypt sensitive data (browser passwords, Wi-Fi keys, app secrets). Stored keys: C:\Users\<Username>\AppData\Roaming\Microsoft\Protect\
How it works: - Generates master keys - Master keys encrypted with the user's Windows login password - Apps use DPAPI to encrypt/decrypt secrets tied to the user account

5. LSASS (Memory-Only Secrets)

Role: - LSASS holds decrypted secrets only while you're logged in, in memory. Examples: - NT hash (active session) - Kerberos tickets - DPAPI master keys - Single Sign-On tokens
Note: Everything in LSASS disappears on reboot.

6. Cached Domain Logins

If using Active Directory (domain): Stored in (registry): HKLM\SECURITY\Cache Protection: - Encrypted using the BootKey - Stored as cached representations (not raw hashes)

Quick Reference:

Windows local passwords -> SAM (encrypted by SAM Key + BootKey)
Browser passwords -> Login Data (SQLite) encrypted with DPAPI
Wi-Fi keys -> WLAN profiles encrypted with DPAPI
App secrets -> DPAPI store, tied to user login
Active session secrets -> LSASS RAM (memory only)
Cached domain logons -> HKLM\SECURITY\Cache (encrypted with BootKey)

Keep this as a quick, corrected reference on Windows credential storage and protection.